

Deja vu? – revisiting M as a big/small-steps semantics

$$\Downarrow_M \subseteq (\mathbb{N} \times \mathbb{N} \times \mathbb{N}) \times \mathbb{N}$$

$$\text{ML1} \frac{}{(i, i, acc) \Downarrow_M acc} \quad \text{ML2} \frac{(i, cnt + 1, 2 * acc) \Downarrow_M p}{(i, cnt, acc) \Downarrow_e p} \quad i \neq cnt$$

$$\rightarrow_M \subseteq (\mathbb{N} \times \mathbb{N} \times \mathbb{N}) \times ((\mathbb{N} \times \mathbb{N} \times \mathbb{N}) \cup \mathbb{N})$$

$$\text{MS1} \frac{}{(i, i, acc) \rightarrow_M acc} \quad \text{MS2} \frac{}{(i, cnt, acc) \rightarrow_M (i, cnt + 1, 2 * acc)} \quad i \neq cnt$$

We want to prove:

- $\forall i, cnt, acc, m \in \mathbb{N}. [(i, cnt, acc) \Downarrow_M m \implies m = 2^{i-cnt} * acc]$
- $\forall i, cnt, acc, m \in \mathbb{N}. [(i, cnt, acc) \rightarrow_M^* m \implies m = 2^{i-cnt} * acc]$